MECH 309: Assignment 4, Question 4

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We consider the linear system $\mathbf{A}\mathbf{x} = \mathbf{b}$ with

$$\mathbf{A} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}, \qquad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \qquad \mathbf{b} = \begin{bmatrix} 4 \\ 3 \\ 2 \\ 1 \end{bmatrix}.$$

Throughout, we use $\mathrm{Im}(\cdot)$ and $\ker(\cdot)$ for the image and kernel.

Question 4(a)

Since $\mathbf{A}$ is already in row echelon form and every column contains a pivot, $\mathbf{A}$ has full column rank. Hence

$$\mathrm{Im}(\mathbf{A}) = \mathrm{span}\{\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3\},$$

where the columns of $\mathbf{A}$ are

$$\mathbf{a}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{a}_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{a}_3 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}.$$

A convenient ("nicer") spanning set is obtained by simple column combinations:

$$\mathbf{a}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \qquad \mathbf{a}_2 - \mathbf{a}_1 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \qquad \mathbf{a}_3 - \mathbf{a}_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}.$$

Therefore,

$$\boxed{\mathrm{Im}(\mathbf{A}) = \mathrm{span}\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix} \right\}}.$$

Since there are three pivot columns,

$$\boxed{\mathrm{rank}(\mathbf{A}) = \dim(\mathrm{Im}(\mathbf{A})) = 3}.$$

Question 4(b)

By rank–nullity for $\mathbf{A} \in \mathbb{R}^{4\times 3}$,

$$\dim(\ker(\mathbf{A})) = 3 - \mathrm{rank}(\mathbf{A}) = 3 - 3 = 0,$$

so $\ker(\mathbf{A}) = \{\mathbf{0}\}$. We can also verify this directly by solving $\mathbf{A}\mathbf{x} = \mathbf{0}$:

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}.$$

From rows 3–4, $x_3 = 0$. Then row 2 gives $x_2 + x_3 = 0 \Rightarrow x_2 = 0$, and row 1 gives $x_1 + x_2 + x_3 = 0 \Rightarrow x_1 = 0$. Hence $\mathbf{x} = \mathbf{0}$ is the only solution, so

$$\boxed{\ker(\mathbf{A}) = \{\mathbf{0}\}, \qquad \text{nullity}(\mathbf{A}) = \dim(\ker(\mathbf{A})) = 0}.$$

Question 4(c)

We have

$$\mathbf{A}^\top = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix} \in \mathbb{R}^{3\times 4}.$$

The columns of $\mathbf{A}^\top$ are

$$\mathbf{c}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \quad \mathbf{c}_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \quad \mathbf{c}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{c}_4 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

Since $\mathbf{c}_4 = \mathbf{c}_3$, we have

$$\text{Im}(\mathbf{A}^\top) = \text{span}\{\mathbf{c}_1, \mathbf{c}_2, \mathbf{c}_3\}.$$

As in part (a), we can "clean" this spanning set by taking simple linear combinations:

$$\mathbf{c}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \mathbf{e}_3, \qquad \mathbf{c}_2 - \mathbf{c}_3 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \mathbf{e}_2, \qquad \mathbf{c}_1 - \mathbf{c}_2 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \mathbf{e}_1.$$

Therefore,

$$\boxed{\text{Im}(\mathbf{A}^\top) = \text{span}\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\} = \mathbb{R}^3}.$$

In particular, $\text{rank}(\mathbf{A}^\top) = 3$ (and thus $\text{rank}(\mathbf{A}^\top) = \text{rank}(\mathbf{A})$):

$$\boxed{\text{rank}(\mathbf{A}^\top) = 3}.$$

Question 4(d)

By rank–nullity for $\mathbf{A}^\top \in \mathbb{R}^{3\times 4}$,

$$\dim(\ker(\mathbf{A}^\top)) = 4 - \text{rank}(\mathbf{A}^\top) = 4 - 3 = 1.$$

To find $\ker(\mathbf{A}^\top)$ explicitly, solve $\mathbf{A}^\top\mathbf{y} = \mathbf{0}$ with

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix}.$$

The system is

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

From the first equation, $y_1 = 0$. From the second, $y_1 + y_2 = 0 \Rightarrow y_2 = 0$. From the third, $y_1 + y_2 + y_3 + y_4 = 0 \Rightarrow y_3 + y_4 = 0$, so $y_4 = -y_3$. Let $y_4 = t$ (free). Then $y_3 = -t$ and

$$\mathbf{y} = t \begin{bmatrix} 0 \\ 0 \\ -1 \\ 1 \end{bmatrix}.$$

Therefore,

$$\boxed{\ker(\mathbf{A}^\top) = \operatorname{span}\left\{ \begin{bmatrix} 0 \\ 0 \\ -1 \\ 1 \end{bmatrix} \right\}, \qquad \operatorname{nullity}(\mathbf{A}^\top) = 1}.$$

Question 4(e)

Let $\mathbf{u} \in \operatorname{Im}(\mathbf{A})$ and $\mathbf{v} \in \ker(\mathbf{A}^\top)$. Using the spanning sets from parts (a) and (d),

$$\mathbf{u} = a \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + b \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} + c \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}, \qquad \mathbf{v} = d \begin{bmatrix} 0 \\ 0 \\ -1 \\ 1 \end{bmatrix}.$$

Then

$$\begin{aligned} \mathbf{u}^\top \mathbf{v} &= d \left(a \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}^\top + b \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}^\top + c \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}^\top \right) \begin{bmatrix} 0 \\ 0 \\ -1 \\ 1 \end{bmatrix} \\ &= d\big(a \cdot 0 + b \cdot 0 + c \cdot \big(1 \cdot (-1) + 1 \cdot 1\big)\big) = 0. \end{aligned}$$

Thus $\mathbf{u}^\top \mathbf{v} = 0$ for all $\mathbf{u} \in \operatorname{Im}(\mathbf{A})$ and $\mathbf{v} \in \ker(\mathbf{A}^\top)$, so

$$\boxed{\operatorname{Im}(\mathbf{A}) \perp \ker(\mathbf{A}^\top)}.$$

Question 4(f)

Compute

$$\mathbf{A}^\top \mathbf{A} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 4 \end{bmatrix}, \qquad \mathbf{A}^\top \mathbf{b} = \begin{bmatrix} 4 \\ 7 \\ 10 \end{bmatrix}.$$

To check whether $\mathbf{A}^\top\mathbf{A}$ is full rank, compute its determinant:

$$\det(\mathbf{A}^\top\mathbf{A}) = \det\begin{bmatrix}1 & 1 & 1\\ 1 & 2 & 2\\ 1 & 2 & 4\end{bmatrix} = 2 \neq 0.$$

Hence $\mathbf{A}^\top\mathbf{A}$ is invertible and therefore full rank:

$$\boxed{\operatorname{rank}(\mathbf{A}^\top\mathbf{A}) = 3}.$$

Question 4(g)

We solve the normal equations

$$\mathbf{A}^\top\mathbf{A}\,\mathbf{x} = \mathbf{A}^\top\mathbf{b}, \qquad \begin{bmatrix}1 & 1 & 1\\ 1 & 2 & 2\\ 1 & 2 & 4\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix} = \begin{bmatrix}4\\ 7\\ 10\end{bmatrix}.$$

Write the augmented matrix and perform Gaussian elimination:

$$\left[\begin{array}{ccc|c}1 & 1 & 1 & 4\\ 1 & 2 & 2 & 7\\ 1 & 2 & 4 & 10\end{array}\right].$$

Step 1: $R_2^{(1)} \leftarrow R_2^{(0)} - 1\cdot R_1^{(0)}, \quad R_3^{(1)} \leftarrow R_3^{(0)} - 1\cdot R_1^{(0)}$:

$$\left[\begin{array}{ccc|c}1 & 1 & 1 & 4\\ 0 & 1 & 1 & 3\\ 0 & 1 & 3 & 6\end{array}\right].$$

Step 2: $R_3^{(2)} \leftarrow R_3^{(1)} - 1\cdot R_2^{(1)}$:

$$\left[\begin{array}{ccc|c}1 & 1 & 1 & 4\\ 0 & 1 & 1 & 3\\ 0 & 0 & 2 & 3\end{array}\right].$$

Thus the upper triangular system $\mathbf{U}\mathbf{x} = \mathbf{c}$ is

$$\mathbf{U} = \begin{bmatrix}1 & 1 & 1\\ 0 & 1 & 1\\ 0 & 0 & 2\end{bmatrix}, \qquad \mathbf{c} = \begin{bmatrix}4\\ 3\\ 3\end{bmatrix}.$$

Backward substitution gives

$$2x_3 = 3 \;\Rightarrow\; x_3 = \frac{3}{2}, \qquad x_2 + x_3 = 3 \;\Rightarrow\; x_2 = \frac{3}{2}, \qquad x_1 + x_2 + x_3 = 4 \;\Rightarrow\; x_1 = 1.$$

Therefore,

$$\boxed{\mathbf{x} = \begin{bmatrix}1\\ \frac{3}{2}\\ \frac{3}{2}\end{bmatrix}}.$$

Question 4(h)

The residual is $\mathbf{r} = \mathbf{b} - \mathbf{A}\mathbf{x}$. First compute

$$\mathbf{A}\mathbf{x} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ \frac{3}{2} \\ \frac{3}{2} \\ \frac{3}{2} \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \\ \frac{3}{2} \\ \frac{3}{2} \end{bmatrix}.$$

Hence

$$\boxed{\mathbf{r} = \mathbf{b} - \mathbf{A}\mathbf{x} = \begin{bmatrix} 4 \\ 3 \\ 2 \\ 1 \end{bmatrix} - \begin{bmatrix} 4 \\ 3 \\ \frac{3}{2} \\ \frac{3}{2} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \frac{1}{2} \\ -\frac{1}{2} \end{bmatrix}}.$$

Question 4(i)

From part (d),

$$\ker(\mathbf{A}^\top) = \operatorname{span}\left\{ \begin{bmatrix} 0 \\ 0 \\ -1 \\ 1 \end{bmatrix} \right\}.$$

We can write

$$\mathbf{r} = \begin{bmatrix} 0 \\ 0 \\ \frac{1}{2} \\ -\frac{1}{2} \end{bmatrix} = \left(-\frac{1}{2}\right) \begin{bmatrix} 0 \\ 0 \\ -1 \\ 1 \end{bmatrix} \in \ker(\mathbf{A}^\top).$$

Thus

$$\boxed{\mathbf{r} \in \ker(\mathbf{A}^\top)}.$$

Question 4(j)

Compute the projection vector $\mathbf{p}$ via $\mathbf{p} = \mathbf{b} - \mathbf{r}$:

$$\boxed{\mathbf{p} = \mathbf{b} - \mathbf{r} = \begin{bmatrix} 4 \\ 3 \\ 2 \\ 1 \end{bmatrix} - \begin{bmatrix} 0 \\ 0 \\ \frac{1}{2} \\ -\frac{1}{2} \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \\ \frac{3}{2} \\ \frac{3}{2} \end{bmatrix}}.$$

Question 4(k)

Using the $\mathbf{x}$ from part (g),

$$\mathbf{A}\mathbf{x} = \begin{bmatrix} 4 \\ 3 \\ \frac{3}{2} \\ \frac{3}{2} \end{bmatrix} = \mathbf{p}.$$

Therefore,

$$\boxed{\mathbf{A}\mathbf{x} = \mathbf{p}}.$$

Question 4(l)

(See attached Strang diagram.) The four fundamental subspaces and their dimensions here are:

$$\mathrm{Im}(\mathbf{A}) \subset \mathbb{R}^4,\ \dim = 3, \qquad \ker(\mathbf{A}^\top) \subset \mathbb{R}^4,\ \dim = 1,$$

$$\mathrm{Im}(\mathbf{A}^\top) = \mathbb{R}^3,\ \dim = 3, \qquad \ker(\mathbf{A}) = \{\mathbf{0}\} \subset \mathbb{R}^3,\ \dim = 0.$$

A convenient basis for $\mathrm{Im}(\mathbf{A}^\top)$ is the canonical basis $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ as shown in part (c). Moreover, $\mathbf{b} = \mathbf{p} + \mathbf{r}$ with $\mathbf{p} \in \mathrm{Im}(\mathbf{A})$ and $\mathbf{r} \in \ker(\mathbf{A}^\top)$, and the least-squares solution is $\mathbf{x} \in \mathbb{R}^3$ satisfying $\mathbf{A}\mathbf{x} = \mathbf{p}$.

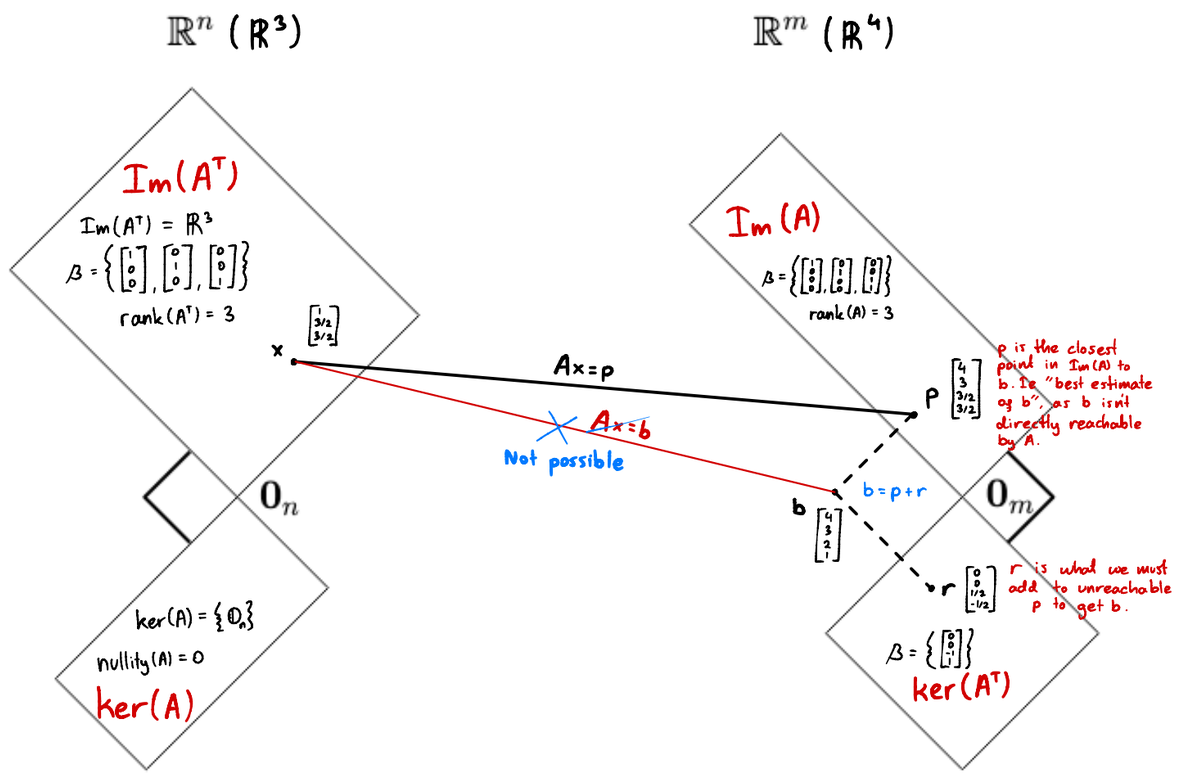

Figure 3: Blank Strang diagram.